

# CAP5735 Data Visualization and Reproducible Research:

## Data Visualization Mini Project 1 - SleepStudy.csv

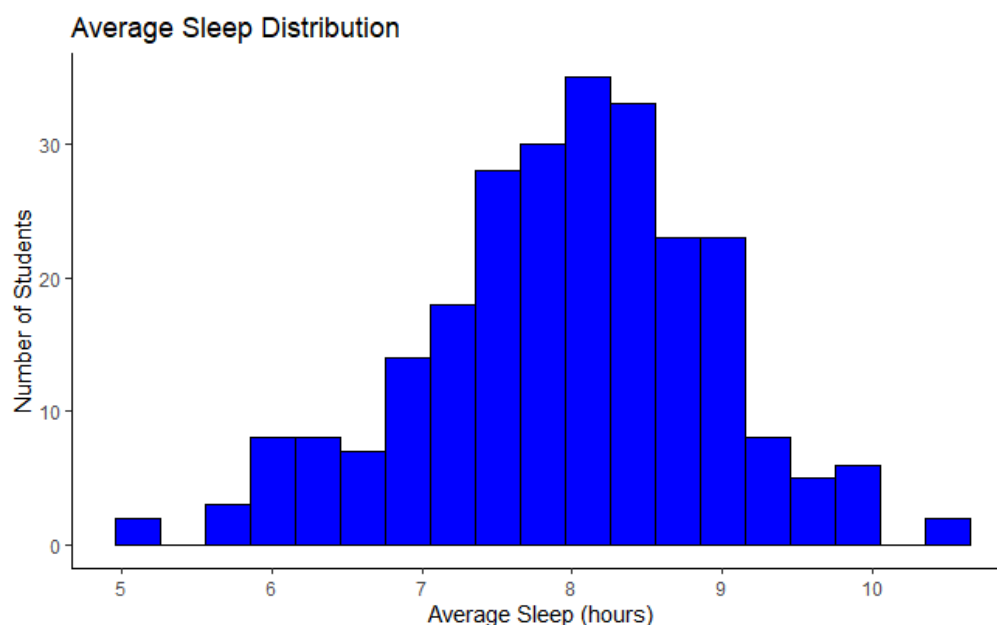
### 1. What were the original charts you planned to create for this assignment?

The original goal was to create a baseline set of visualizations to understand the basic structure of the dataset. The planned charts included:

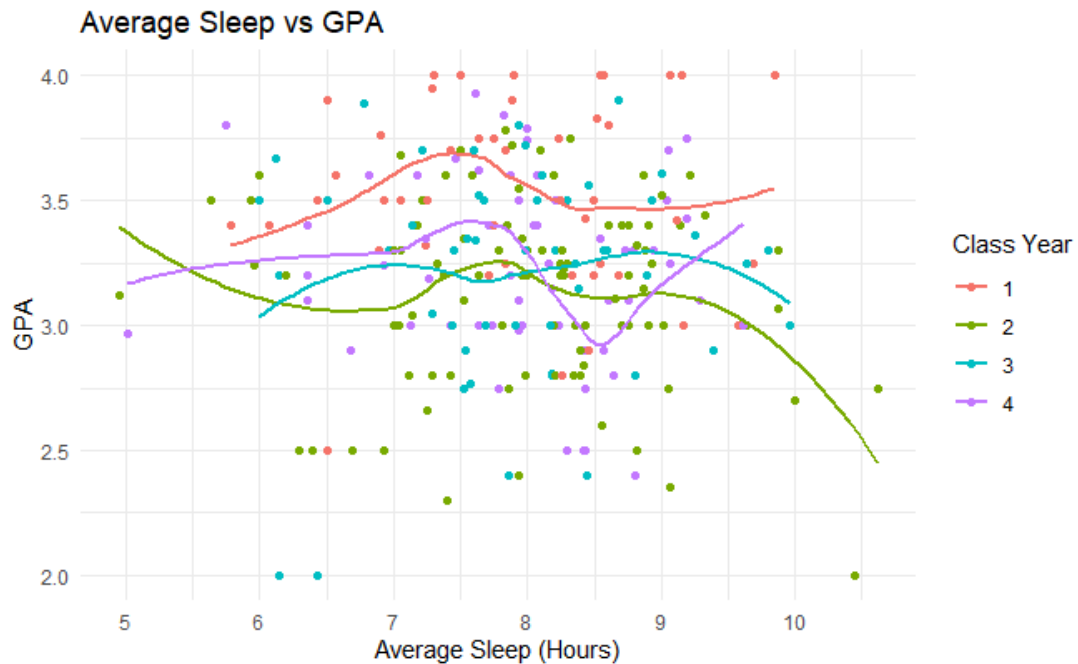
- A basic histogram to gauge the distribution of average sleep.
- Boxplots to see if categorical traits (like being an "Early Bird" vs. "Night Owl") impacted numerical outcomes (like GPA).
- Scatter plots with basic trend lines to explore linear relationships, such as stress versus sleep quality.

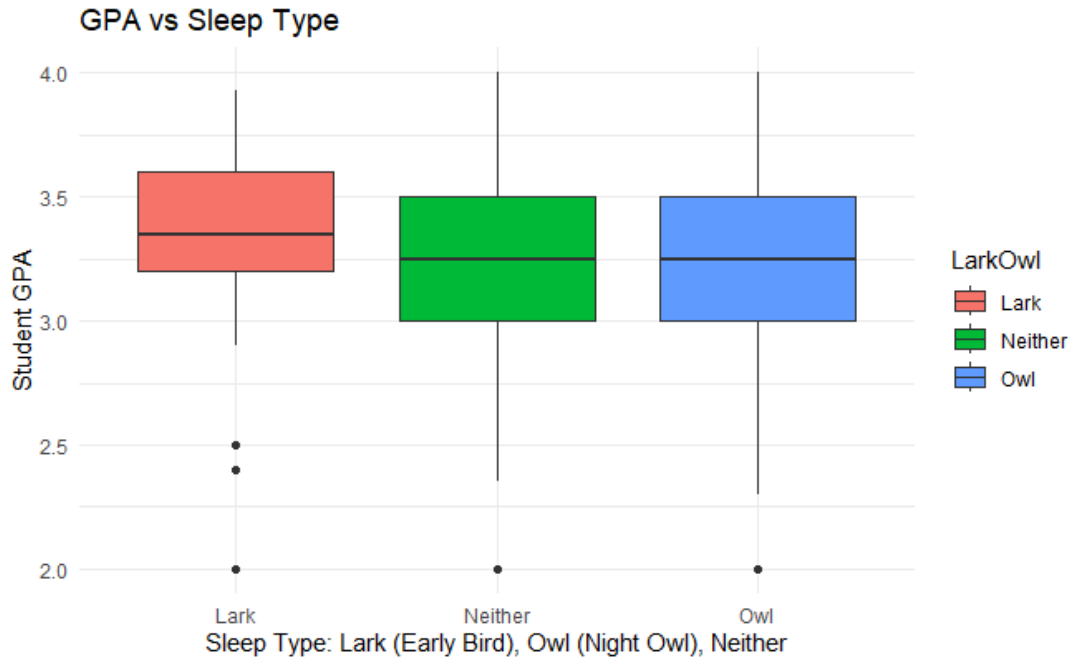
### 2. What story could you tell with your plots?

- **Histogram:** The Average Sleep Distribution histogram set a baseline, and allowed me to make sure the overall data made sense visually. This was important as sleep was the primary factor of this dataset.

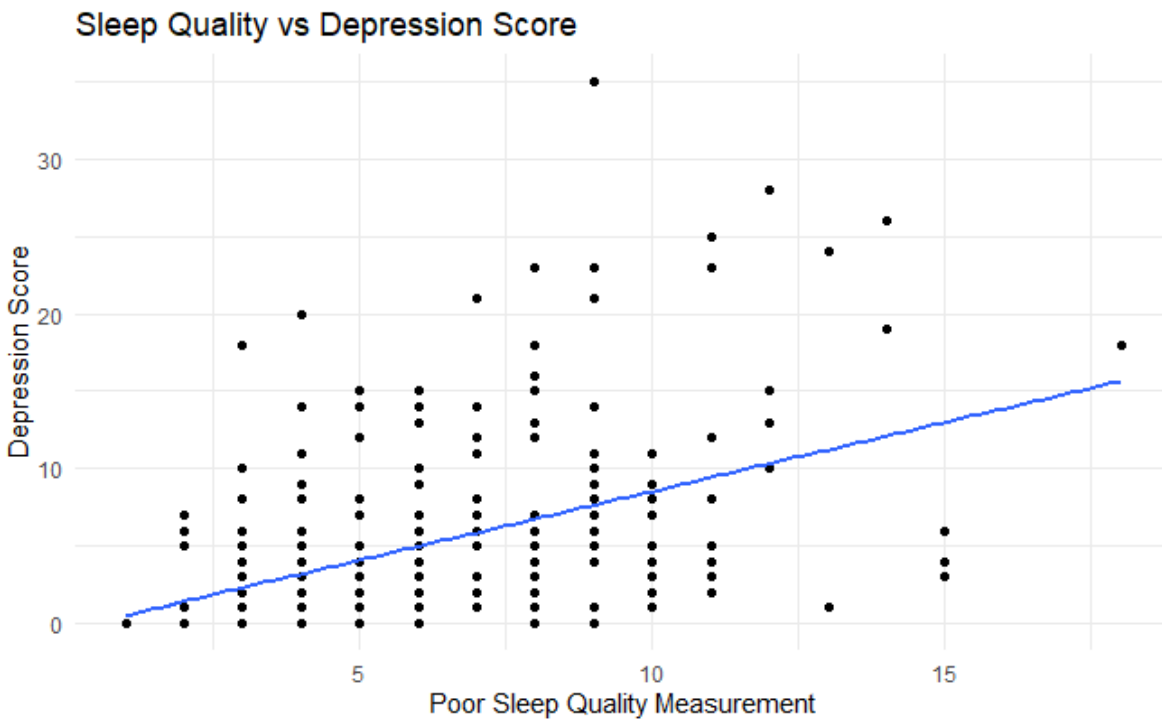


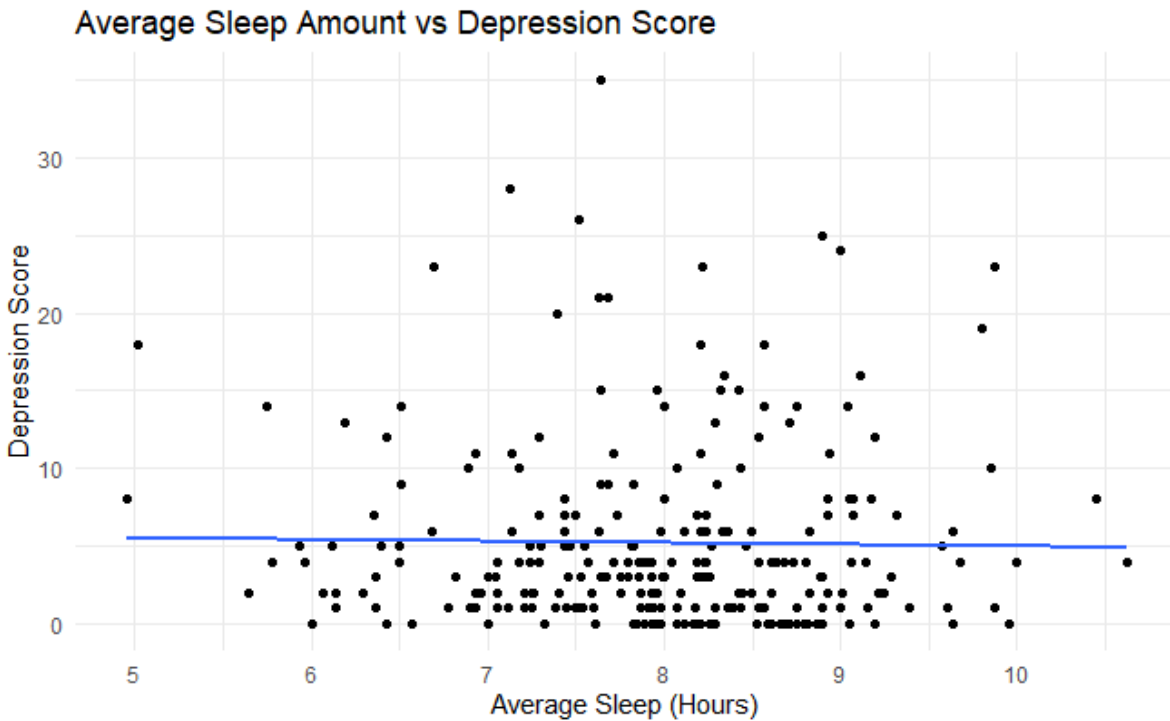
- **Sleep vs GPA:** The scatter plot of Average Sleep vs. GPA reveals how sleep duration correlates with grades across different class years. The smoothing curve helped analyze where the benefits of sleep stopped and the consequences of too much sleep began. 7.5 hours seemed to be the overall sweet spot, but each grade level was slightly different. The GPA vs. Sleep Type boxplot showed how students' overall sleep habits result in academic advantages or disadvantages. Overall, Larks (early birds) had a higher average GPA, while night owls and students who didn't fall into either category were virtually identical.



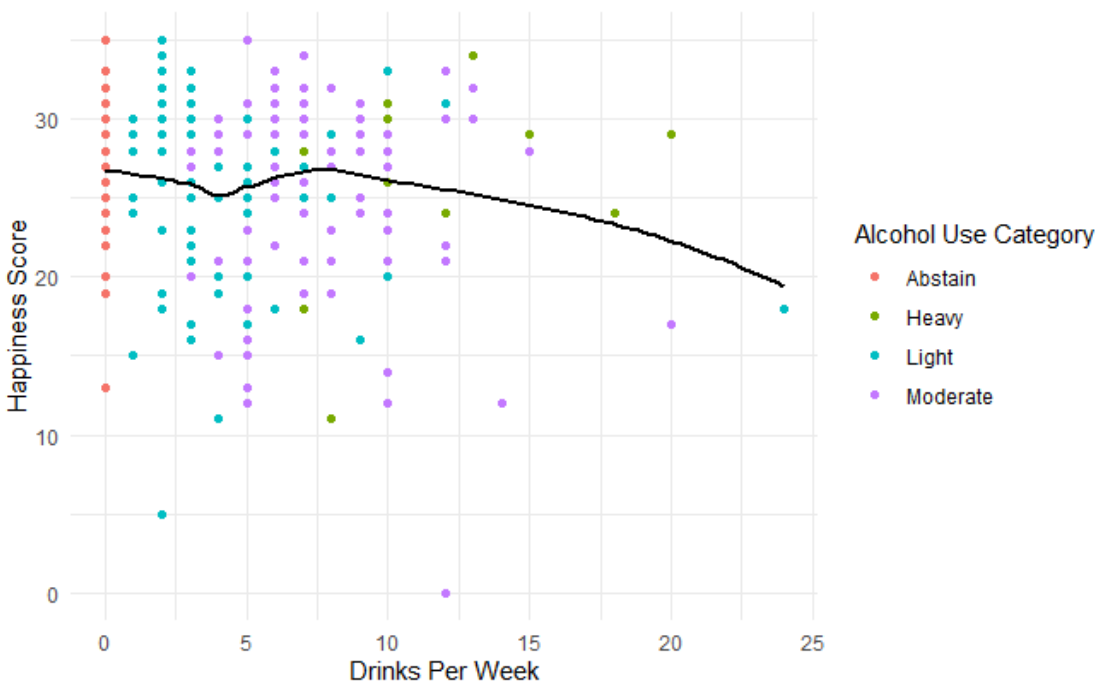


- **Sleep and Depression:** The two side-by-side depression scatter plots tell a compelling story about mental well-being. By plotting Depression Score against both Poor Sleep Quality and Average Sleep Amount, it shows that sleep quality has a large impact on depression, however, sleep amount is unrelated which was surprising to me.





- **Drinking vs Happiness:** Lastly, I looked at drinking rates and happiness because I was interested in those variables while exploring the dataset and assumed they would be negatively correlated. Plotting them showed what I expected with a slight unexpected dip in happiness around 4 drinks, which went back up before continuing the expected negative correlation at about 7 drinks.



Note: Due to the lack of documentation in the dataset I assumed the drinks variable was per week as the values ranged from 0-25. I was also unsure as to how the sleep quality and depression score variables were calculated/measured but the findings were still strong.

### **3. How did you apply the principles of data visualizations and design for this assignment?**

**Chart Selection** - Using the appropriate chart was important to show the data in the easiest way to understand

**Use of Color** - I utilized colors for when data had trends that varied within the groups like the class year for sleep vs gpa. However, I also chose not to use color on every graph as the ones without it didn't need it and therefore it would have only cluttered the graph more.

**Clean Themes and Trends** - Utilizing the theme\_minimal and classic helped clean up the graph backgrounds while still allowing for the data to be easily read. I also used geom\_smooth to create trend lines when necessary to showcase trends or lack thereof, allowing for conclusions to be made quicker.